

Homework 6

2.1.1 Let f, g, h be integrable functions on the line. Prove the following:

- a. The Fourier transform of $(x_1, x_2) \mapsto f(x_1 - x_2)g(x_2)$ is $\widehat{f}(\xi_1) \cdot \widehat{g}(\xi_1 + \xi_2)$.

Proof. The Fourier transform for this function is

$$\begin{aligned}
 & \int_{\mathbb{R}^2} f(x_1 - x_2)g(x_2)e^{-2\pi i(x_1, x_2) \cdot (\xi_1, \xi_2)} d(x_1, x_2) \\
 &= \int_{\mathbb{R}} \int_{\mathbb{R}} f(x_1 - x_2)g(x_2)e^{-2\pi i x_1 \xi_1} \cdot e^{-2\pi i x_2 \xi_2} dx_1 dx_2 \\
 &= \int_{\mathbb{R}} g(x_2)e^{-2\pi i x_2 \xi_2} \int_{\mathbb{R}} f(x_1 - x_2)e^{-2\pi i x_1 \xi_1} dx_1 dx_2 \\
 &= \int_{\mathbb{R}} g(x_2)e^{-2\pi i x_2 \xi_2} \int_{\mathbb{R}} f(x_1 - x_2)e^{-2\pi i(x_1 - x_2)\xi_1} \cdot e^{-2\pi i x_2 \xi_1} dx_1 dx_2 \\
 &= \int_{\mathbb{R}} g(x_2)e^{-2\pi i x_2(\xi_1 + \xi_2)} \int_{\mathbb{R}} f(x_1 - x_2)e^{-2\pi i(x_1 - x_2)\xi_1} dx_1 dx_2 \\
 &= \widehat{f}(\xi_1) \int_{\mathbb{R}} g(x_2)e^{-2\pi i x_2(\xi_1 + \xi_2)} dx_2 = \widehat{f}(\xi_1) \cdot \widehat{g}(\xi_1 + \xi_2).
 \end{aligned}$$

□

Note: this can of course be done the same way as parts b and c. However, it was not until after I had written this out in LaTeX that I realized the technique to solve b and c. It seems unnecessary to write the other technique in LaTeX.

- b. The Fourier transform of $k : (x_1, x_2) \mapsto f(x_1 + x_2)g(x_1 - x_2)$ is $\frac{1}{2}\widehat{f}(\frac{\xi_1 + \xi_2}{2}) \cdot \widehat{g}(\frac{\xi_1 - \xi_2}{2})$.

Proof. We use the 8th identity from proposition 2.1.6, that $\widehat{f \circ A} = \frac{1}{|\det A|} \widehat{f} \circ (A^t)^{-1}$.

Consider the function $h : (x_1, x_2) \mapsto f(x_1)g(x_2)$, and the following matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Then $A^t = A$ and $|\det A| = 2$. Notice that $k(x_1, x_2) = h \circ A \cdot (x_1, x_2)$. By remark 2.1.3 from the text, $\widehat{h}(\xi_1, \xi_2) = \widehat{f}(\xi_1) \cdot \widehat{g}(\xi_2)$. So, we have that

$$\begin{aligned}
 \widehat{k}(\xi_1, \xi_2) &= \widehat{h \circ A}(\xi_1, \xi_2) = \frac{1}{2} \widehat{h} \circ A^{-1} \cdot (\xi_1, \xi_2) \\
 &= \frac{1}{2} \widehat{h} \circ \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \cdot (\xi_1, \xi_2) \\
 &= \frac{1}{2} (\widehat{f} \cdot \widehat{g})\left(\frac{\xi_1 + \xi_2}{2}, \frac{\xi_1 - \xi_2}{2}\right) \\
 &= \frac{1}{2} \widehat{f}\left(\frac{\xi_1 + \xi_2}{2}\right) \cdot \widehat{g}\left(\frac{\xi_1 - \xi_2}{2}\right).
 \end{aligned}$$

□

c. The Fourier transform of $(x_1, x_2, x_3) \mapsto f(x_1 - x_2 - x_3)g(x_2 + x_3)h(x_3)$ on $\mathbb{R}^3$ is

$$(\xi_1, \xi_2, \xi_3) \mapsto \widehat{f}(\xi_1) \cdot \widehat{g}(\xi_1 + \xi_2) \cdot \widehat{h}(\xi_3 - \xi_2).$$

Proof. Define $h^* : (x_1, x_2, x_3) \mapsto f(x_1)g(x_2)h(x_3)$. Consider the matrices defined below:

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad A^t = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, \quad (A^t)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \quad |\det A| = 1$$

So, we have

$$\begin{aligned} \widehat{h^* \circ A}(\xi_1, \xi_2, \xi_3) &= (\widehat{h^*} \circ \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix})(\xi_1, \xi_2, \xi_3) \\ &= \widehat{h^*}(\xi_1, \xi_1 + \xi_2, \xi_3 - \xi_2) \\ &= \widehat{f}(\xi_1) \widehat{g}(\xi_1 + \xi_2) \widehat{h}(\xi_3 - \xi_2). \end{aligned}$$

□